

# Manual

## Split and Merged Operations BDE-SSG 8.2

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# 1 Overview: Split and Merged Operations

## Purpose

This function package provides functions that can be used to split or consolidate operations and also the corresponding posting functions.

## Implementation notes

You use the function package if:

- You would like to combine different short-running operations in order to be able to enter them "one-to-one".

or if

- You want to shorten operation lead times or better utilize workplaces / machines available for production by processing operations in parallel.

## Integration

The functions contained in this package are also integrated in different function groups of the Shop Floor Data Collection module. Emphasis here is on data entry, i.e. posting at the terminal, and posting in the system.

## Features

- Split operations
  - User dialog for generating and managing split operations
  - Supply of split operations for MES planning and entry in order to plan split operations in parallel and to be able to post quantities and times to them.
  - Merging production activities to the original operation and uploading to the ERP
- Merged operations
  - Function in the MOC for combining operations with short processing duration into merged operations
  - User dialog at the Windows terminal for generating, posting and managing merged operations
  - Supply of merged operations for MES planning and entry
  - Posting times and quantities proportionally to original operations based on configuration and confirmations/uploads

## 2 Splitting Operations

|  |   |  |
|--|---|--|
|  | → |  |
|--|---|--|

### Purpose

You use this function if you want to shorten the lead time of operations or improve utilization of several workplaces/ machines available for production.

### Integration

This function splits an operation into two or more operations. An operation can be split using the following functions:

- [Order sequencing.](#)
- [Edit operations.](#)
- [Graphic planning:](#) can be called up via the context menu of the operation.

Data is always uploaded to the PPS/ ERP system according to the order/ operation number of the original operation.

### Requirements

An operation can only be split if the following conditions are met:

- The length of the split number is set to 1 (maximum: 9 splits) or 2 (maximum: 99 splits) in the HYDRA basic parameter settings.
- The identifier "May be split" is set at the operation.
- The operation to be split cannot be a merged operation.
- The operation is not a split operation. It is not allowed to split a split operation once more.
- The operation has not yet started, i.e. it must be marked with status (control indicator) "prepared".

### Splitting an operation

|                        |          |
|------------------------|----------|
| Function authorization | op.split |
|------------------------|----------|



The function "Split operation"  allows you to split the total production quantity of an operation into several operations. The number of operations subjected to splitting is predefined as per the number of splits configured in the order backlog. If the indicator "May be split" is not set for the operation, you cannot split the operation.

The system runs through the following processing steps when splitting an operation:

### Checks

- A check is run to verify whether the operation may be split. If the indicator "May be split" is set for the operation, you can split the operation.
- The number of splits entered must be greater than 1.
- The entered number of splits must be less than the maximum number of splits defined at the operation.
- The operation to be split may not have already been split (split master).
- The operation to be split may itself not be an operation split.
- The operation to be split may not be a merged operation.

If these conditions are met, the operation may be split.

### Processing

The system issues the numbers for the splits automatically by distributing a consecutive number. The number is shown in the field "split": The remainder of the number (order/ operation) matches the one of the original operation ("split master"). The number of splits depends on the length/number of characters defined for splits in the HYDRA basic parameter settings.

Example:

Operation 47100100 0100 is divided up into to three splits; the number length for the split defined in the basic settings is 2.

Results of splitting:

Split master: 47100100 0100 (can no longer be logged on)

Individual splits: 47100100 0100 01

47100100 0100 02

47100100 0100 03

When an operation is split, the following values of the original operation are distributed to the splits based on the number of splits:

- Target quantity (primary, secondary, tertiary, base quantity unit)
- Target scrap (primary, secondary, tertiary, base quantity unit)
- Processing time

The quantities are generally divided up in full numbers. If they cannot be distributed evenly, the remainder is allocated to the last operation split.

All other data is transferred from the original operation.

After splitting, the order is flagged for lead time scheduling.



After splitting, only the generated operation splits can be posted. The original operation (split master) is marked as *locked* and cannot be posted. It is no longer shown in the backlog of orders.

The HYDRA Shop Floor Scheduling module no longer shows the split master and, therefore, the master cannot be planned any longer, but only its split operations.

Changes to operation splits, in particular changes in planning (e.g. changes to planned dates or changes to the planned workplace) do not affect the split master.

Please keep in mind that after an operation has been split, any new changes from the ERP system to this original operation ("split master") will have no effect on the individual operations split off.

If the target quantity update is enabled, the operation splits will not be updated, but only the split master.

## Cancel operation splits

|                        |                 |
|------------------------|-----------------|
| Function authorization | op.splitrelease |
|------------------------|-----------------|



When an operation split is canceled, all of the operations split off are deleted and only the original operation, which up until now was the split master, continues to exist. The split master is now treated as a normal operation.

Operation splits can only be canceled if all splits of the operation still show a status "prepared".

## Displaying operation splits

Whether split operations are displayed depends on the application:

- The original operation (split master) is not shown in graphic planning.
- The split master is shown in the following applications:
  - Edit operations
  - Order information

Whether the single operation splits are displayed also depends on the application:

- The [graphic planning](#) generally shows operation splits. In contrast to this, the split master is no longer shown after splitting.
- The single operations split off are shown in the following applications, provided that the relevant option has been selected:
  - [Edit operations](#)
  - [Order information](#)

## Posting to operation splits

Where split operations are concerned, the log record is ONLY generated for the operation split; no log record is generated for the split master itself.

Only the split master's status is updated if the following events occur:

- interrupt operation
- log off operation

## Uploading to the PPS/ ERP system

The values recorded or posted to operation splits are generally uploaded to the PPS/ ERP system according to the split master.

## Splitting of operations subject to batch management

### General

The "split operation" function may also be run for operations subject to management in batches. The function is identical to the split function of conventional operations.

### Procedure

The user applies the "split OP" function and splits off an OP from the "split master" in the given operation quantity (primary quantity).

The operation is split. Data relating to material is transferred from the split master to the operations split off.

The required quantity is calculated subject to the input quantity and the quantity split off.

### Results after splitting the operation:



With regard to the material, the following data/ values are accepted or used by the operations split off:

- Order header data
  - Finished article (number of the finished article) --> taken over from split master
  - Article name --> taken over from split master
  - Batch number --> taken over from split master (for the batch being generated in this order, e.g. transferred from the ERP; identical for all operation splits)
  - Material type of the finished article --> taken over from split master
- OP data
  - Material number of output material/article --> taken over from split master
  - Material type of output material/article --> taken over from split master
  - Name of output material/article --> taken over from split master
  - Indicator "batch management requirement" --> taken over from split master
  - Indicator "serial numbers required" --> taken over from split master
- Component list
  - All data included in the component list is copied or transferred from the split master.
  - The required quantity is recalculated from the OP split off based on the quantity split off.
- Output batch change:
  - When output batches are changed, the number of the split master is added to the output batch.
- Uploads to the ERP system:
  - Uploads to the ERP system are always performed in relation to the split master.

## 3 Enhanced Split Function

### Purpose

You use the split function if you would like to shorten the lead time of operations or you would like to improve the capacity utilization of several workplaces/machines available for production.

In contrast to the simple split function, the enhanced split function provides the following features:

- You can split an operation into split operations with different target quantities and setup times.
- You can split an operation that has already been split (split operation) into further split operations with different target quantities and setup times.
- You can delete individual splits.

### Integration

This function splits an operation into two or more operations. An operation can be split using the following functions:

- [Order sequencing](#).
- [Edit operations](#).
- [Graphic Planning](#): can be called via the context menu of the operation.

Data is always uploaded to the PPS/ERP system according to the order/operation number of the original operation.

### Requirements

An operation can only be split if the following conditions are met:

- The length of the split number is set to 1 (maximum 9 splits) or 2 (maximum 99 splits) in the [Basic settings](#).
- In the [Basic settings](#), you must set the option "Enhanced split function" to "J - Yes".
- The option "May be split" is set for the operation and the field "Max. number of splits" includes a value greater than 2.
- The operation to be split cannot be a merged operation.

## Split operation

|                        |          |
|------------------------|----------|
| Function authorization | op.split |
|------------------------|----------|



You can split an operation using the function "split operation".

To perform splits, select the respective operation and start the split function using the context menu or by clicking the respective button. A dialog opens where the planner may define into how many operations the operation is to be split and what quantities are planned for each split operation.

### Field description of "split operation"

The dialog shows different information depending on whether the operation is split for the first time or if splitting is performed for an operation that has already been split.

In the field descriptions below, it is specified for each field if the field is available when the OP is

- (1) split for the first time or
- (2) when the OP is further split.

#### MES order number (1) (2)

Operation that you want to split. In case of a first split, it is the operation that may be split (split master); for the following splits, it is the split master.

#### Target quantity (P) (1) (2)

Target quantity and unit of the split master according to the specifications made in the pool of orders. The target quantity and the other quantities generally refer to the primary quantity of the operation.

#### Actual quantity (1) (2)

(Yield) quantity that has been uploaded/confirmed so far. This quantity is entered for the split operations. The quantity is also used to update the split master status.

It includes the actual quantity of operations that have already been finished as well as the actual quantity of running or interrupted operations.

#### Remaining quantity for split OPs (1) (2)

For all split OPs that have not yet been finished (split OPs with control indicator S, V, L or U) with the difference resulting from "target quantity minus actual quantity" being > 0: sum total of remaining quantities.

Operations with a difference between target quantity and actual quantity that is less than 0 are not used. This difference results from overproduction. Such a surplus is no longer produced and, as a result, no longer used for further splitting.

**Difference (1) (2)**

This quantity results from:

"Target quantity" (P) minus "actual quantity" minus "remaining quantity of split OPs"

If this quantity is unequal 0, the text "DIFFERENCE" is shown in red. You can therefore easily identify whether/when the target quantity of the split master is entirely distributed.

**Max. number of splits (1) (2)**

The maximum number of splits defined for the split master.

**Previous number of splits (1) (2)**

Number of splits that have been performed so far. This number specifies the total number of split operations and the splits that must be performed (splits included in the list)

**Remaining run time (1) (2)**

Remaining run time of the split master according to the remaining run time formula.

**Setup time (1) (2)**

Target setup time of the split master according to the specifications made in the pool of orders. The setup time is taken over into the list during splitting and can be adjusted there individually for each split.

The information listed above generally refers to the operation to be split or to the split master, provided that the operation has already been split.

If the operation has already been split, i.e. it is now the split master, the below-mentioned information is shown for the split operation that is used for further splitting. If the operation is split for the first time, these fields are *not* displayed:

**Split OP (2)**

Split operation that is to be split or that is to be used for further splitting.

**Target quantity (P) (2)**

Target quantity of the split operation.

**Actual quantity (2)**

(Yield) quantity uploaded/confirmed so far for this split operation

## Interactive input options of the planner

If you enter the "number of splits" and, if required, the quantity that you want to split, and if you then press the "split" button, the splits are generated in the list. You can still change the target quantity and the setup time.

### Number of splits (1) (2)

Planners can now specify how many splits they want to create.

By default, the field is assigned the number of splits that are still possible (maximum number of splits minus current number of splits).

### Quantity to be split (1) (2)

In addition, the planner can define a split quantity. In this case, the individual splits are assigned this quantity in the table.

If the quantity to be split is 0, the remaining quantity is distributed evenly among the splits. However, the quantity may still be changed for each split.

You can add splits until the maximum number of splits of the split master is reached.

### Generate OP with difference quantity (1)

This checkbox is only visible if the OP has not yet been split. The option is therefore not set by default.

If the checkbox is enabled, another split operation is shown in the table in addition to the split OPs. Condition: the total target quantity of the split OPs to be generated (included in the list) is less than the remaining quantity of the operation to be split.

For this operation, you can change all target data, except the target quantity itself. The target quantity cannot be changed, because it is automatically calculated: remaining quantity calculated using the quantity of the operation and the total of all target quantities of the split OPs to be generated.

### Set off target qty. of the split OP (2)

This checkbox is only visible if the OP is a split operation. The option is therefore not set by default.

If this checkbox is set, the target quantity of the split OP is updated, i.e. the target quantity of the split OP is reduced by the total target quantity of the split operations to be generated. However, this is only the case if this sum does not exceed the remaining quantity of the split operation (the remaining quantity results from the difference between target quantity and actual quantity). Otherwise, you cannot enable the checkbox.

## Table columns

### Split (1) (2)

The system generates the future split number. You cannot change the number.

The next free number is used as split number, i.e. it is checked which number is currently the highest split number.

**Target quantity (P) (1) (2)**

Target quantity of the split operation. The planner may change the target quantity, provided it is not the split generated due to the option "Generate OP with difference quantity".

The target quantity of the new entries results from the "quantity to be split".

**Setup time (1) (2)**

Target setup time of the split operation. The planner may change the setup time.

For new entries, the setup time of the split master is preassigned.

**Remaining run time (1) (2)**

Remaining run time (remaining processing time) according to the remaining run time formula that is defined for the split master taking into account the target quantity of the split operation. HYDRA calculates this remaining run time and it cannot be changed by the planner.

The totals row shows:

- Column "Split": number of entries,
- Column "Target quantity": Total of the target quantity,
- Column "Setup time": Total of setup times,
- Column "Remaining run time": Total of remaining run times.

An error message is output if the entered number of splits (existing and new splits) exceeds the maximum number of splits.

It is allowed to enter an (admissible) number of splits and a split quantity that is greater than the target quantity of the operation. However, a respective information saying "difference" is displayed for the user.

**Editing of split entries**

Existing entries can be edited or deleted using the "edit" or "delete" button. Multiple selections are supported for deletion.

Once entries have been confirmed by clicking OK, they are updated in the relevant dialog.

The target quantity cannot be changed if the checkbox "Generate OP with difference quantity" is set.

**Split result**

The entries are checked, once the dialog "split operation" has been confirmed by clicking OK. Error messages are shown if necessary:

- Splits having a target quantity 0 are not allowed.

- In general, the maximum number of splits is checked against the split master.

Then the system generates the split operation(s). They are directly generated in the database.

One or several new operations are generated by splitting. Their ID differs from the original split master by the split number. There are the following operations, for example:

- 123456789012 0100 = Split master
- 123456789012 0100 1 = Split operation 1
- 123456789012 0100 2 = Split operation 2

## Additional information



After splitting, you can only make postings for the generated operation splits. The original operation (split master) is marked as *locked* and no postings can be made for this operation. It is no longer shown in the backlog of orders.

The HYDRA Shop Floor Scheduling module no longer shows the split master and, therefore, the master cannot be planned any longer, but only its split operations.

Changes to operation splits, in particular changes in planning (e.g. changes to planned dates or changes to the planned workplace) do not affect the split master.

Please keep in mind that after an operation has been split, any new changes from the ERP system to this original operation ("split master") will have no effect on the individual operations split off.

If the target quantity update is enabled, the operation splits will not be updated, but only the split master.

## Cancel operation splits

|                        |                 |
|------------------------|-----------------|
| Function authorization | op.splitrelease |
|------------------------|-----------------|



A split operation can be canceled by clicking the button , if the split operation has not yet been started.

In contrast to the simple split function, the enhanced split function only cancels, i.e. deletes the selected split operation. The split master and the other split operations remain unchanged.



Please note: If a split operation is deleted, its target quantity will not be transferred to other, still existing split operations.

If the last split operation is deleted, the split master becomes a "normal" operation.

## Displaying operation splits

Whether or not split operations are displayed depends on the context of the relevant application.

Split master:

- The original operation (split master) is not shown in graphic planning.
- The split master is shown in the following applications:
  - Edit operations
  - Order information

Split operations:

- The [Graphic planning](#) generally shows operation splits. In contrast to this, the split master is no longer shown after splitting.
- The single operations split off are shown in the following applications, provided that the relevant option has been selected:
  - [Edit operations](#)
  - [Order information](#)

## Posting for split operations

Where split operations are concerned, the log record is ONLY generated for the split operation; no log record is generated for the split master itself.

Only the status of the split master is updated. The following events trigger an update:

- interrupt operation
- log off operation



## **Uploading part quantities to the PPS/ERP system**

The values recorded or posted for split operations are generally uploaded to the PPS/ERP system with reference to the split master.

## 4 Processing of Merged Operations

### Purpose

Merged operations are a special form of the serial production. In the relevant planning level (e.g. in HYDRA shop floor scheduling) or on the shop floor terminal, you combine different operations with a short run time each to a logic group, which then has a manageable run time. This group of operations is called "merged operation". The system creates a "substitute" operation representing the merged operation. You then plan and log on only the "substitute" operation that contains all single operations. You use different configurable perspectives to distribute the data recorded.

### Integration

To create a merged operation, HYDRA provides two methods:

#### Creating merged operations on the terminal

The operator creates the merged operation on the terminal. When you log on a merged operation, you enter the single operations one after the other and assign them to the relevant person. When you log off or interrupt this merged operation, you must only enter the person.

#### Creating merged operations on the MOC

The function *Generate merged operation* combines separate operations and builds a substitute operation that stands in for all the other short operations. You log on the substitute operation on the terminal.

If you have created a merged operation on the MOC, you can schedule this merged operation using the MOC planning functions. This is not possible with merged operations created on the terminal.

### Requirements

You cannot create merged operations on the terminal and on the MOC. This is incompatible. You must configure in the [HYDRA basic settings](#) which one of the two methods is used. Subject to this configuration, the functions for merged operations are available either on the terminal or on the MOC only.

### 4.1 Creating merged operations on the terminal

Using the function *Merged operation* on the terminal, you can combine several operations and build one merged operation (MOP) on the terminal.

Order processing using merged operations is useful if:

- several operations with short run times are combined to build one operation (lower posting efforts),
- several operations are produced using one workplace at the same time. The system cannot make timely logon/logoff postings for the separate operations (for example during hardening in the metal industry or smoking in the food industry).

### 4.1.1 Different posting types of merged operations

The below-mentioned types of postings for merged operations are specified in the [Terminal configuration](#).

#### "Generation per person"

When you log on a MOP, you log on the combination of person and orders. The person who combines several operations to build a MOP is automatically logged on to all separate operations included in the MOP.

Only one MOP is possible per person. If a person performs another MOP logon at a later time, then these separate operations are automatically assigned to the already existing MOP.

A MOP can only be logged on, logged off or interrupted. You cannot log on additional persons to a person-related MOP.

#### "Generation per machine"

With this configuration, you can use the function *Log on merged operation* on the terminal to combine several OPs and build one MOP. Per machine, you can create one MOP.

Just as it is the case for single operations, several persons can log on to this MOP. If you interrupt or log off the MOP, all persons logged on are automatically logged off. The posting of quantities and the distribution of times is made in accordance with the specifications described in the sections that follow.

### 4.1.2 Posting of merged operations on the terminal

For information on how to create merged operations on the terminal, refer to the relevant terminal documentation (*for Windows or DOS terminals*).

Merged operations are created on the terminal as described in the relevant documentation. The terminal generates a substitute order number for the merged operation. Depending on the posting type, the user's badge number or the machine/workplace number is integrated in this order number:

#### Merged operation number for the "generation per person" option

Depending on the length of the order number and staff badge number configured in the system, the number of the merged operation is created as follows:

The length of the order no. is at least four characters longer than the staff badge no.:

**SAM-XXXXX**                      xxxxx represents the staff badge no.

The length of the order no. is at least two characters longer than the staff badge no.:

**S-XXXXX**                      xxxxx represents the staff badge no.

The order number length is **not** at least two characters longer than the staff badge number:



In this case, the standard function "merged operations on the terminal" cannot be used.

### Merged operation number for the "generation per machine" option

Depending on the length of the order number and the machine number configured in the system, the number of the merged operation is created as follows:

Length of the order number is at least two characters longer than the length of the machine/workplace number:

**S-XXXXX**                      xxxxx represents the machine/workplace number

Length of the order number is at least four characters longer than the length of the machine/workplace number:

**SAM-XXXXX**                      xxxxx represents the machine/workplace number

The length of the order number is **not** at least two characters longer than the length of the machine/workplace number:



In this case, the standard function "merged operations on the terminal" cannot be used.

## 4.1.3 Booking of merged operations created on the terminal

The actual times recorded for the MOP can be distributed according to the following methods:

According to the **standard times**, i.e. in relation to the standard times of the separate OPs

According to the **default quantities**, i.e. in relation to the target quantities of the separate OPs

According to the **individual OP**, i.e. according to the number of the logged on single OPs

The posting method is defined in the [Basic settings](#).



The options "according to standard time" or "according to default quantity" must not be used together with the option *Proportionate RPA posting in personnel postings* in the basic settings.



Please contact MPDV Support to configure the MOP posting method.

## Example - Distribution according to standard time

**Distribution of the actual times of the MOP according to the standard times of the separate OPs:**

3 operations are combined and form one MOP:

| OP   | Standard time |
|------|---------------|
| OP01 | 8000 sec      |
| OP02 | 600 sec       |
| OP03 | 6000 sec      |

A log record for the MOP with the run time of 12000 sec must now be distributed to the separate OPs.

The following formula is used to distribute the times

$$\text{Posted times} = \frac{\text{Total duration of the log record} * \text{standard time of individual OP}}{\text{Total of the standard times of all OPs}}$$

The following values result for the separate OPs:

| OP   | Run time booked |
|------|-----------------|
| OP01 | 6575 sec        |
| OP02 | 493 sec         |
| OP03 | 4932 sec        |

Sample calculation for OP 01:

Sum of standard times of all separate OPs = 8000 + 600 + 6000 = 14600

Run time booked = 12000 \* 8000/14600 = 6575 sec

### PLEASE NOTE:

The standard time of an operation is calculated using the **target setup time + target duration**.

## Example - Distribution according to standard time

**Distribution of the actual times of the MOP according to the target quantities of the separate OPs:**

3 operations are combined to form one MOP:

| OP   | Target quantity |
|------|-----------------|
| OP01 | 1000            |
| OP02 | 500             |
| OP03 | 2500            |

A log record for the merged OP with the run time 12000 sec must now be booked to the different OPs. To distribute the times, the following formula applies

$$\text{Posted time} = \frac{\text{Total duration of log record} * \text{target quantity of individual OP}}{\text{Total of target quantities of all individual OPs}}$$

The following values result for the separate OPs:

| OP   | Run time booked |
|------|-----------------|
| OP01 | 3000 sec        |
| OP02 | 1500 sec        |
| OP03 | 7500 sec        |

Sample calculation for OP01:

Sum of target quantities of all separate OPs = 1000 + 500 + 2500 = 4000

Run time booked = 12000 \* 1000/4000 = 3000 sec

## Example – Distribution according to the number of separate operations

### Distribution of actual times of the MOP according to single OPs

3 OPs are combined to form one MOP:

A log record for the MOP with the run time 12000 sec must now be distributed to the different OPs. To distribute the times, the following formula applies

$$\text{Posted time} = \frac{\text{Total duration of the log record}}{\text{number of individual OPs}}$$

The following values result for the separate OPs:

| OP   | Run time booked |
|------|-----------------|
| OP01 | 4000 sec        |
| OP02 | 4000 sec        |
| OP03 | 4000 sec        |

The **quantities and the times recorded** are also **posted for the person** who processes the MOP.

## 4.1.4 Notes on the configuration and the processing

### Configuration and use

- By default, the “merged operation” function is not active and must be enabled for a specific terminal in the terminal configuration.
- Only the ADE terminal provides MOP functions on DOS terminals CT56x und CT73x . On Windows terminals, the MOP functions can also be used in "MDE" operation mode.
- To produce a MOP, the option *Logon of several OPs* must be set in the workplace/machine configuration, i.e. it is permitted to log on several operations at this workplace.
- When you create merged operations on the terminal, the option *Max. OPs per person* of the HR master data is used. If the option *Max. OPs per person* is set to =1 for the person, the person can only log on one operation as merged operation.
- A person can use a maximum of 20 separate operations to build a merged operation.
- When you log on a MOP on a DOS terminal Typ CT56x und CT73x , a sequencing list is not supported, i.e. you must enter the order numbers via barcode or manually.
- If the number of pieces is not recorded for a merged operation, the actual quantity produced is set to the target number of pieces of the separate OPs when the MOP is logged off.
- If the quantity is recorded for a merged operation, the quantity entered for each operation is posted to the different operations when the merged operation is interrupted or logged off.
- For merged operations that are logged on to workplaces/machines with automatic recording of quantities, the proportionate quantity is posted to each separate operation. The quantities are distributed according to the same key as it is the case for times. See the sections above.
- When the MOP is logged off/interrupted, you must only enter the badge number (MOP person). All operations included in a person's merged operation are logged off automatically.

### Supported automatic functions/data collection options

- Terminate OP when reaching target quantity, only option "Y" (as of SP10/2016)  
When the MOP is interrupted, the system checks if the single OP has reached its target quantity. If yes, the relevant OP is finished instead of interrupted. The system does not automatically log off a single OP from a merged operation, if the target quantity of a single OP is reached or exceeded with an upload of a partial quantity or an automatic recording of quantities.
- Terminate OP instead of interrupting it  
When you interrupt the MOP, the system checks if the single OP is finished and not interrupted.
- Interrupt OP instead of terminating it  
When you log off the MOP, the system checks if the single OP is interrupted and not finished.
- Automatic release of succeeding OP

- Configurable posting behavior with change/end/beginning of shift  
The order type specifies if the behavior is configured for the machine or the person. Special feature:  
With person-related MOPs, the merged operation is interrupted when the person is logged off, regardless of the OP setting.

### **Not supported or restricted automatic functions/data collection options**

- Terminate OP when reaching target quantity: options U, F and K
- Option *Proportionate posting of machine time with parallel OPs* in the machine configuration because this option performs the proportionate posting that is configured for the MOP.
- Target quantity reached output (is only performed for the total quantity of the MOP)
- Collection of serial number
- Data collection with batch management requirement
- Posting of resources
- With the milestone processing, you must be aware that a single OP of a MOP cannot be unmerged from the MOP when a posting is made for the preceding or succeeding OP.

Example:

When you log off an OP, the preceding OP cannot be logged off automatically if the preceding OP is included in a merged OP (person or machine).

The MOP must be logged off manually.

- Other restrictions exist with the CAQ integration and the data collection for MOPs, for example the automatic logon and logoff of inspection OPs.

### **Waiting period processing**

If you must postdate a MOP logon because of the waiting period processing, the first single operation of this merged operation is postdated. Further OPs that might be added to the MOP at a later time need not be postdated.

If a clock-out is posted in the PZE for the person that processed the MOP, then all included operations are automatically interrupted.

If the waiting period processing function is configured accordingly, it is also possible that the person who processes the MOP is automatically logged on again as soon as they clock-in in HYDRA-PZE.

## **4.2 Creating merged operations on the MOC**

You can use this function to combine separate operations to build merged operations on the MOC.

The functions to create and cancel merged operations are available in the *Order overview* and in the *Order sequencing* dialog (to call the order sequencing, the BDE-FST must be licensed).



If a merged operation is created on the MOC, then the "members" of a merged operation are no longer displayed. Also in the sequencing list on the terminals, the member operations are no longer displayed.

The number of members in a merged operation is not limited. You can only add prepared orders/OPs to merged operations that are not already contained in other merged operations.

You **cannot** integrate merged operations themselves into other merged operations.

All postings of a merged operation on the terminal are made for all "members" of a merged operation. You can log on, interrupt and log off merged operations on the terminal exactly like normal operations.

## 4.2.1 Creating merged operations on the MOC

The method to create merged operations is described [here](#).

## 4.2.2 Booking of merged operations (created on the MOC)

To distribute the quantities and times recorded for a merged operation among the included operations, the system supports the "homogeneous" and the "inhomogeneous" method for the merged operations that were not created on the MOC.

### Homogeneous merged operations

With a homogeneous merged operation, quantities and times are distributed using the overrun principle: all separate operations are "filled up" one after the other according to the specified target number of pieces. If the production surpasses the quantity planned for the total merged operation, then this excess and the respective times are added to the single operation having the largest order number.

Simplified example:

| OP   | Target quantity | Target run time |
|------|-----------------|-----------------|
| OP01 | 200             | 4.0 hours       |
| OP02 | 450             | 9.0 hours       |
| OP03 | 250             | 4.5 hours       |

A log record of the merged operation containing the real quantity 500 and real duration 5.0 hours must now be redistributed among the separate operations. The number of pieces is posted to the separate OPs one after the other (up to the specified target quantity). Depending on the posted number of pieces, the following formula is used to calculate the times:

$$\text{Posted time} = \frac{\text{Total log record time} * \text{posted quantity}}{\text{Total quantity of the log record}}$$

The following values result for the separate OPs:

| OP          | Posted quantity | Run time booked |
|-------------|-----------------|-----------------|
| Single OP01 | 200             | 2.0 hours       |
| Single OP02 | 300             | 3.0 hours       |
| Single OP03 | 0               | 0.0 hours       |

## Non-homogeneous merged operations

With an **inhomogeneous merged operation**, the quantities and times are posted proportionately to the separate operations included. If excess production takes place, then the excess is distributed among the different operations.

Simplified example:

| OP          | Target quantity | Standard time |
|-------------|-----------------|---------------|
| Single OP01 | 400             | 8.0 hours     |
| Single OP02 | 20              | 0.5 hours     |
| Single OP03 | 30              | 1.0 hours     |

A log record for the merged operation including an actual quantity 200 and a run time 4 hours must now be redistributed to the separate operations. The following formulas apply:

### Distribution of times

$$\text{Posted time} = \left( \frac{\text{Standard time of individual OP} *}{\text{Sum total of all individual OPs}} \right) * \text{Total duration of log record}$$

### Distribution of quantities

$$\text{Posted quantity} = \left( \frac{\text{Target quantity of individual OP}}{\text{Sum total of all individual OPs}} \right) * \text{Actual quantity of log record}$$

The following values result for the separate OPs:

| OP   | Posted quantity | Run time booked |
|------|-----------------|-----------------|
| OP01 | 178             | 3.4 hours       |
| OP02 | 9               | 0.2 hours       |
| OP03 | 13              | 0.4 hours       |

Sample calculation (for single OP01)

Run time booked =  $8/9.5 * 4 = 3.4$

Quantity booked =  $400/450 * 200 = 178$  pieces.



The standard time of an operation is calculated using the **target setup time + processing time**. All data required for this calculation is contained in the order backlog of the operation.



If uneven values like 2.666 result when the piece number is distributed, the calculated quantity is cut off and 2 pieces are booked for the operation. **Exception:** With the last operation, the difference between quantity posted and quantity already booked is used.

The type used for the merged operation in the entire system is specified once in the [HYDRA basic settings](#) using the BDE option *Process merged operations*.

The upload of the **times recorded** to the PPS system is performed according to the definition of the merged operation type – homogeneous or inhomogeneous.

If a member of a merged operation is finished e.g. via ERP interface, then this operation is skipped during the calculations described above.

## 4.2.3 Further notes

### Validation checks

Validation checks, which are performed during the booking process (e.g. target quantity validation for upload of a part quantity), are only performed for the merged operation and not for the separate operations assigned to the merged operation.

### Changing OP data

#### Changing the MOP

If you make changes to the merged operation, the single OPs included are NOT changed.

#### Changing a member of a merged operation

If you make changes to one of the operations included in a merged operation, the master operation is not changed.

Whether you can make changes or not depends on the relevant status of the operation (same process as with a normal operation).



## 5 Generating and canceling a merged operation

### Overview

|                        |                                                                                                                                                                                 |                                                      |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Menu                   | Production control → Production overview → Operations<br>Production control → Production support → Pool of orders<br>Production control → Production support → Order sequencing |                                                      |
| Function authorization | op.colopcreate<br>op.coloprelease                                                                                                                                               | Generate merged operation<br>Cancel merged operation |

### Purpose

You use the function *Generate merged operation* to combine separate operations and to build one substitute OP. On the terminal, you then log on the substitute operation.

### Integration

If you have created a merged operation on the MOC, you can schedule this merged operation using the MOC planning functions. This is not possible with merged operations created on the terminal.

For information on the processing and booking of merged operations, see [here](#).

### Requirements

To generate merged operations on the MOC, you must have activated the relevant configuration in the *HYDRA basic settings*. Subject to this configuration, the functions for merged operations are available either on the terminal or on the MOC only.

### Generate merged operation

To generate a merged operation, select all the operations that should be part of the merged operation. To select several operations (or to de-select them), hold the CTRL key down and click on the separate operations.



Click the button  to call the function *Generate merged operation*. The dialog *Generate merged operation* opens. The following input fields are available:

#### MOP order number

Enter the order number for the merged operation. If you leave this field empty, HYDRA automatically assigns the number.

#### Operation

By default, the operation number 0000 is preassigned here.

**Target quantity (P)**

For all selected operations, the system calculates for each operation:

Calculated quantity = target quantity (P) minus yield quantity (P).

The calculated quantities are totaled if the result is > 0.

The system uses the sum total to populate the field *Target quantity (P)*.

Note: the units are not converted; all operations must have the same primary quantity unit.

**Target scrap (P)**

For all selected operations, the system calculates for each operation:

Calculated quantity = target scrap (P) minus (actual) scrap (P).

The calculated quantities are totaled if the result is > 0.

The system uses the sum total to populate the field *Target scrap (P)*.

Note: the units are not converted; all operations must have the same primary quantity unit.

**Setup time**

By default, the maximum setup time of the selected operations is used.

**Processing time**

By default, the total of all processing times of the separate operations is used as processing time.

**Reference OP**

The data of the order/operation stored in field *Reference OP* is used for the merged operation ("copy template"). The system uses the operation having the lowest order/operation number of the selected operations to populate this field.

Confirm the dialog to generate the new merged operation.



You can combine a maximum of 50 separate operations to build a merged operation.

If you change the data of the merged operation, the inventory data of the OPs used to build this merged operation is NOT changed.

When you cancel the merged operation, the data of the included operations is still the same than at the time when the merged operation was created.

**Cancel merged operation**

As long as a merged operation has not been started (V status), you can cancel this merged operation and

restore the separate operations included by clicking the button . Once a merged operation has been started, you cannot cancel the merge anymore.



If you change the data of the merged operation, the inventory data of the OPs used to build this merged operation is NOT changed.

When you cancel the merged operation, the data of the included operations is still the same than at the time when the merged operation was created.

## Display merged operations

With evaluations/overviews, you can control via selection if the evaluation/overview integrates merged operations or the operations included in merged operations.

### **Indiv. OPs**

Only "normal" operations are shown, no merged operations or operations included in merged operations.

### **Merged operations**

Only merged operations (MOP master) are shown.

### **Indiv. OPs summarized in merged OPs**

Only the operations included in MOPs are shown.

In the application *Operations* (transaction code op), you can show the column *MOP*. This column shows the MES order number of the merged operation to which the single operation is assigned or the MES order number of the merged operation itself.